1.2 Cube Roots

Name _____ Period ____ Date _____

Algebra I 1.2 · Student sheet

I can…

- find the cube root of a perfect cube.
- evaluate expressions that contain cube roots.
- use cube roots to solve real-life problems.

Vocabulary · fill in the missing word

Perfect cube — a number that is the cube of an _____, such as $27 = 3^3$

Cube root — $\sqrt[3]{a}$ is the number that, when _____, gives a

Negative cube roots — the cube root of a negative number is negative: $\sqrt[3]{-8}$ = _____

Part 1 · Perfect cube table

n	1	2	3	4	5	6	7	8	9	10
n^3										

Use the table: $\sqrt[3]{27}$ = _____ $\sqrt[3]{-125}$ = _____ $\sqrt[3]{1000}$ = _____ $\sqrt[3]{(1/8)}$ = _____

Can a cube root be negative? Why? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Cube root: The cube root of a is the number that, when _____, equals a.

Negative cube roots: $\sqrt[3]{(-8)}$ = _____ because (____)$^3 = -8$.

Example 1

Find each cube root.

a. $\sqrt[3]{64}$ b. $\sqrt[3]{-27}$ c. $\sqrt[3]{\left(\frac{8}{125}\right)}$

Solving $x^3 = a$: To solve $x^3 = a$, take the _____ of both sides.

Example 2

A cube has a volume of 343 cubic inches. What is the length of each edge?

Example 3

Evaluate $2\sqrt[3]{x} - 5$ when $x = 216$.

Example 4

Which is greater, $\sqrt[3]{100}$ or 5?

Part 3 · Partner practice

Partner 1

Find each cube root.

a. $\sqrt[3]{-1000}$ b. $\sqrt[3]{\left(\frac{1}{64}\right)}$

Partner 2

Evaluate $10 - \sqrt[3]{x}$ when $x = -8$.